

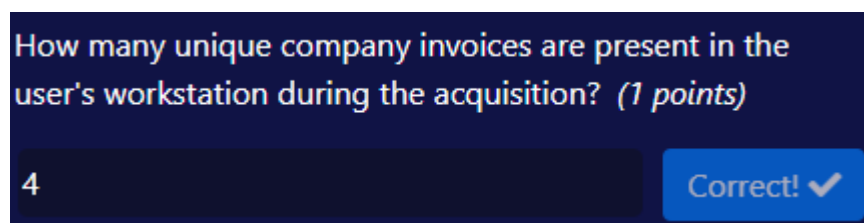
Scenario

While NAE Systems is developing a new product, threat intelligence sources report that sensitive product data has been leaked and is now being advertised for sale on the dark web. In response, the Incident Response team begins an internal investigation and traces suspicious outbound network activity to an employee workstation potentially linked to the breach. Initial findings suggest that the employee may have accessed the material out of curiosity, but investigators must determine whether that curiosity escalated into unauthorized possession, staging, transfer, or exfiltration of the compromised file. As part of the investigation, you are tasked with determining whether the employee possessed a copy of the file, collecting forensic evidence to support that finding, and identifying any suspicious activity on the host that may indicate unauthorized access, file staging, execution, transfer, or exfiltration.-----Zimmerman tools/ EZ tools

Note:

- **MFTECmd.exe** — parses the raw `$J` stream into a clean CSV with columns like `Name` , `UpdateReasons` , `EntryNumber` , and timestamps.
- **\$J** : Every NTFS volume keeps a running log of file-system-level events: creations, renames, deletions, extensions, security changes. That log lives at `'Extend$UsnJrnl'` , inside a data stream called `'Extend$UsnJrnl'` , inside a data stream called `$J`. Think of it as a flight recorder for the file system — it doesn't store file *contents*, it stores file *behavior*.
 - `$J` itself causes problems in PowerShell because `$` is the variable sigil. You need a **backtick** escape (``$J`) or the shell will try to interpret it as a variable.

Investigation



By searching the "invoice", we get many result including pdf and ink files. The pdf file is the readable and common invoice file format. So let's type 'pdf' in the filter area under the Name field. The result include many same items, so we have to dedupe. Group by the field "Name". We get 4 items which is the final answer.

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File Tools Tabs View Help

output.csv

Name

Extension	Entry Number	Sequence Number	Parent Entry Number	Parent Sequence Number	Parent Path	Update Sequence Number	Update Timestamp	Update Reasons
> Name: invoice-apr.pdf (Count: 19)								
> Name: invoice-feb.pdf (Count: 19)								
> Name: invoice-jan.pdf (Count: 19)								
> Name: invoice-mar.pdf (Count: 17)								

☒ Name ☐ Contains ☐ pdf

C:\Users\BTLO\Test\Desktop\output.csv

Total lines 332,257 Visible lines 74 Open files: 1 Search options

How many unique company files ending in .docx extension are present in the user's workstation that were owned by the company? (1 points)

7

Correct! ✓

Notice :

- When a .docx file is opened, a \$<filename>.docx file created. It is lock file which is not a real docx file. So those \$<filename>.docx should not be counted to the answer.
- .lnk is the shortcut file. So .lnk files should not be counted into the answer.

output.csv

Name ▲ ▼

	Extension	Entry Number	Sequence Number	Pa
▼	•	=	=	=
	> Name: \$I3M0T8D.docx (Count: 14)			
	> Name: \$IQH54V7.docx (Count: 26)			
	> Name: \$R3M0T8D.docx (Count: 4)			
	> Name: \$RQH54V7.docx (Count: 4)			
▶	> Name: nae-receipts.docx (Count: 7)			
	> Name: nae-receipts.docx.lnk (Count: 13)			
	> Name: nae-report2025Dec.docx (Count: 17)			
	> Name: nae-report2025Dec.docx.lnk (Count: 24)			
	> Name: nae-report2026April.docx (Count: 27)			
	> Name: nae-report2026Feb.docx (Count: 27)			
	> Name: nae-report2026Jan.docx (Count: 7)			
	> Name: nae-report2026Jan.docx.lnk (Count: 13)			
	> Name: nae-report2026Mar.docx (Count: 7)			
	> Name: nae-report2026Mar.docx.lnk (Count: 13)			
	> Name: nae-report2026May.docx (Count: 17)			
	> Name: nae-report2026May.docx.lnk (Count: 13)			

◀

x [x] Name Contains .docx

Filter Editor

And

Name	Contains	.docx
Name	Does not contain	lnk
Name	Does not contain	\$

`Contains([Name], '.docx') And Not Contains([Name], 'lnk') And Not Contains([Name], '$')`

OK Cancel Apply

> Name: nae-receipts.docx (Count: 7)
> Name: nae-report2025Dec.docx (Count: 17)
> Name: nae-report2026April.docx (Count: 27)
> Name: nae-report2026Feb.docx (Count: 27)
> Name: nae-report2026Jan.docx (Count: 7)
> Name: nae-report2026Mar.docx (Count: 7)
> Name: nae-report2026May.docx (Count: 17)

The company was working on an unreleased home automation project, and it believed it had been stolen. What was the name of this product? (2 points)

homepilot

Correct! ✓

search “home” , and group extension to figure out how many file types are related. Zip type is the fittest type for a project.

Name	Entry Number	Sequence Number	Parent Entry Number	Parent Sequence Number	Parent Path	Update Sequence Number	Update Timestamp	Update Reason
Extension: .css (Count: 3)								
Extension: .evtx (Count: 2)								
Extension: .html (Count: 222)								
Extension: .jpeg (Count: 7)								
Extension: .js (Count: 222)								
Extension: .lnk (Count: 37)								
Extension: .propertymap (Count: 219)								
Extension: .stringsmap (Count: 219)								
Extension: .variablemap (Count: 219)								
Extension: .zip (Count: 58)								
home\pilot-main.zip	148165	13	1904	24		1765803976	2026-04-15 08:51:04	RenameNewName
home\pilot-main.zip	148165	13	1904	24		1765804072	2026-04-15 08:51:04	RenameNewName Close
home\pilot-main.zip	148165	13	1904	24		1765804168	2026-04-15 08:51:07	StreamChange
home\pilot-main.zip	148165	13	1904	24		1765804264	2026-04-15 08:51:07	NamedDataExtend StreamChange
home\pilot-main.zip	148165	13	1904	24		1765804360	2026-04-15 08:51:07	NamedDataExtend StreamChange Close
home\pilot-main.zip	30497	16	4460	16		1765809680	2026-04-15 08:51:19	FileCreate
home\pilot-main.zip	30497	16	4460	16		1765809776	2026-04-15 08:51:19	FileCreate SecurityChange
home\pilot-main.zip	30497	16	4460	16		1765809872	2026-04-15 08:51:19	DataExtend FileCreate SecurityChange
home\pilot-main.zip	30497	16	4460	16		1765809968	2026-04-15 08:51:19	DataOverwrite DataExtend FileCreate SecurityChange
home\pilot-main.zip	30497	16	4460	16		1765810064	2026-04-15 08:51:19	DataOverwrite DataExtend FileCreate SecurityChange StreamChange
home\pilot-main.zip	30497	16	4460	16		1765810176	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataExtend FileCreate SecurityChange StreamChange
home\pilot-main.zip	30497	16	4460	16		1765810272	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange StreamChange
home\pilot-main.zip	30497	16	4460	16		1765810368	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange
home\pilot-main.zip	30497	16	4460	16		1765810464	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange Close
home\pilot-main.zip	30497	16	4460	16		1765811072	2026-04-15 08:51:30	RenameOldName
home\pilot-main.zip	30497	16	4460	16		1765811424	2026-04-15 08:51:30	RenameNewName Close
home\pilot-v1.zip	30497	16	4460	16		1765811520	2026-04-15 08:51:30	RenameOldName
home\pilot-v1.zip	148165	13	1904	24		1765813660	2026-04-15 08:53:20	ObjectIdChange
home\pilot-v1.zip	30497	16	4460	16		1765835792	2026-04-15 08:53:20	ObjectIdChange Close
home\pilot-main.zip	148248	19	1904	24		1765990400	2026-04-15 09:45:07	RenameNewName
home\pilot-main.zip	148248	19	1904	24		1765990496	2026-04-15 09:45:07	RenameNewName Close
home\pilot-main.zip	148248	19	1904	24		1765990592	2026-04-15 09:45:09	StreamChange
home\pilot-main.zip	148248	19	1904	24		1765990688	2026-04-15 09:45:09	NamedDataExtend StreamChange
home\pilot-main.zip	148248	19	1904	24		1765990784	2026-04-15 09:45:09	NamedDataExtend StreamChange Close
home\pilot-main.zip	148215	23	4460	16		1765992848	2026-04-15 09:45:18	FileCreate
home\pilot-main.zip	148215	23	4460	16		1765992944	2026-04-15 09:45:18	FileCreate SecurityChange
home\pilot-main.zip	148215	23	4460	16		1765993040	2026-04-15 09:45:18	DataExtend FileCreate SecurityChange
home\pilot-main.zip	148215	23	4460	16		1765993136	2026-04-15 09:45:18	DataOverwrite DataExtend FileCreate SecurityChange
home\pilot-main.zip	148215	23	4460	16		1765993232	2026-04-15 09:45:18	DataOverwrite DataExtend FileCreate SecurityChange StreamChange

Following Q3, the user renamed the files into their respective versions. What are the new file names? (2 points)

home\pilot-v1, home\pilot-v2

Correct! ✓

At 08:51:04 the first time the file presents, the file names home\pilot-main.zip, then rename behaviors recored in the "Update Reasons" . Home\pilot-v1 and Home\pilot-v2 present.

Name	Entry Number	Seq...	Parent Entry...	Parent...	Parent...	Update Seq...	Update Timestamp	Update Reasons
Extension: .zip (Count: 58)								
home\pilot-main.zip	148165	13	1904	24		1765803976	2026-04-15 08:51:04	RenameNewName
home\pilot-main.zip	148165	13	1904	24		1765804072	2026-04-15 08:51:04	RenameNewName Close
home\pilot-main.zip	148165	13	1904	24		1765804168	2026-04-15 08:51:07	StreamChange
home\pilot-main.zip	148165	13	1904	24		1765804264	2026-04-15 08:51:07	NamedDataExtend StreamChange
home\pilot-main.zip	148165	13	1904	24		1765804360	2026-04-15 08:51:07	NamedDataExtend StreamChange Close
home\pilot-main.zip	30497	16	4460	16		1765809680	2026-04-15 08:51:19	FileCreate
home\pilot-main.zip	30497	16	4460	16		1765809776	2026-04-15 08:51:19	FileCreate SecurityChange
home\pilot-main.zip	30497	16	4460	16		1765809872	2026-04-15 08:51:19	DataExtend FileCreate SecurityChange
home\pilot-main.zip	30497	16	4460	16		1765809968	2026-04-15 08:51:19	DataOverwrite DataExtend FileCreate SecurityChange
home\pilot-main.zip	30497	16	4460	16		1765810064	2026-04-15 08:51:19	DataOverwrite DataExtend FileCreate SecurityChange StreamChange
home\pilot-main.zip	30497	16	4460	16		1765810176	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataExtend FileCreate SecurityChange StreamChange
home\pilot-main.zip	30497	16	4460	16		1765810272	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange StreamChange
home\pilot-main.zip	30497	16	4460	16		1765810368	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange
home\pilot-main.zip	30497	16	4460	16		1765810464	2026-04-15 08:51:19	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange Close
home\pilot-main.zip	30497	16	4460	16		1765811072	2026-04-15 08:51:30	RenameOldName
home\pilot-main.zip	30497	16	4460	16		1765811424	2026-04-15 08:51:30	RenameNewName Close
home\pilot-v1.zip	30497	16	4460	16		1765811520	2026-04-15 08:51:30	RenameOldName
home\pilot-v1.zip	148165	13	1904	24		1765813660	2026-04-15 08:53:20	ObjectIdChange
home\pilot-v1.zip	30497	16	4460	16		1765835792	2026-04-15 08:53:20	ObjectIdChange Close
home\pilot-main.zip	148248	19	1904	24		1765990400	2026-04-15 09:45:07	RenameNewName
home\pilot-main.zip	148248	19	1904	24		1765990496	2026-04-15 09:45:07	RenameNewName Close
home\pilot-main.zip	148248	19	1904	24		1765990592	2026-04-15 09:45:09	StreamChange
home\pilot-main.zip	148248	19	1904	24		1765990688	2026-04-15 09:45:09	NamedDataExtend StreamChange
home\pilot-main.zip	148248	19	1904	24		1765990784	2026-04-15 09:45:09	NamedDataExtend StreamChange Close
home\pilot-main.zip	148215	23	4460	16		1765992848	2026-04-15 09:45:18	FileCreate
home\pilot-main.zip	148215	23	4460	16		1765992944	2026-04-15 09:45:18	FileCreate SecurityChange
home\pilot-main.zip	148215	23	4460	16		1765993040	2026-04-15 09:45:18	DataExtend FileCreate SecurityChange
home\pilot-main.zip	148215	23	4460	16		1765993136	2026-04-15 09:45:18	DataOverwrite DataExtend FileCreate SecurityChange
home\pilot-main.zip	148215	23	4460	16		1765993232	2026-04-15 09:45:18	DataOverwrite DataExtend FileCreate SecurityChange StreamChange

Following Q4, what are the entry numbers of the files after being renamed? (1 points)

148227, 148280

Correct! ✓

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File Tools Tabs View Help

output.csv

Name homeplot x Find

Ext...	Entry Number	Sequ...	Parent...	Parent...	Update Sequen...	Update Timestamp	Update Reasons
.zip							create
.zip	148215	23	4460	16	1765993288	2026-04-15 09:45:18	FileCreate
.zip	148215	23	4460	16	1765992944	2026-04-15 09:45:18	FileCreate SecurityChange
.zip	148215	23	4460	16	1765993040	2026-04-15 09:45:18	DataExtend FileCreate SecurityChange
.zip	148215	23	4460	16	1765993136	2026-04-15 09:45:18	DataOverwrite DataExtend FileCreate SecurityChange
.zip	148215	23	4460	16	1765993232	2026-04-15 09:45:18	DataOverwrite DataExtend FileCreate SecurityChange StreamChange
.zip	148215	23	4460	16	1765993328	2026-04-15 09:45:18	DataOverwrite DataExtend NamedDataExtend FileCreate SecurityChange StreamChange
.zip	148215	23	4460	16	1765993424	2026-04-15 09:45:18	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange StreamChange
.zip	148215	23	4460	16	1765993520	2026-04-15 09:45:18	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange StreamChange
.zip	148215	23	4460	16	1765993616	2026-04-15 09:45:18	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange StreamChange Close
Name: homeplot-v1.zip (Count: 9)							
.zip	148277	13	92961	31	1766173600	2026-04-15 10:50:31	FileCreate
.zip	148277	13	92961	31	1766173696	2026-04-15 10:50:31	FileCreate SecurityChange
.zip	148277	13	92961	31	1766173792	2026-04-15 10:50:31	DataExtend FileCreate SecurityChange
.zip	148277	13	92961	31	1766173888	2026-04-15 10:50:31	DataOverwrite DataExtend FileCreate SecurityChange
.zip	148277	13	92961	31	1766173984	2026-04-15 10:50:31	DataOverwrite DataExtend FileCreate SecurityChange StreamChange
.zip	148277	13	92961	31	1766174080	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataExtend FileCreate SecurityChange StreamChange
.zip	148277	13	92961	31	1766174176	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange StreamChange
.zip	148277	13	92961	31	1766174272	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange StreamChange
.zip	148277	13	92961	31	1766174368	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange StreamChange Close
Name: homeplot-v2.zip (Count: 9)							
.zip	148280	15	92961	31	1766174464	2026-04-15 10:50:31	FileCreate
.zip	148280	15	92961	31	1766174560	2026-04-15 10:50:31	FileCreate SecurityChange
.zip	148280	15	92961	31	1766174720	2026-04-15 10:50:31	DataExtend FileCreate SecurityChange
.zip	148280	15	92961	31	1766174816	2026-04-15 10:50:31	DataOverwrite DataExtend FileCreate SecurityChange
.zip	148280	15	92961	31	1766174912	2026-04-15 10:50:31	DataOverwrite DataExtend FileCreate SecurityChange StreamChange
.zip	148280	15	92961	31	1766175008	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataExtend FileCreate SecurityChange StreamChange
.zip	148280	15	92961	31	1766175104	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange StreamChange
.zip	148280	15	92961	31	1766175200	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange StreamChange
.zip	148280	15	92961	31	1766175296	2026-04-15 10:50:31	DataOverwrite DataExtend NamedDataOverwrite NamedDataExtend FileCreate SecurityChange BasicInfoChange StreamChange Close

Extension Contains .zip And Update Reasons Contains create

When was the first version of the home automation product downloaded and written to disk, as per the \$J record? (2 points)

2026-04-15 08:51:19

Correct! ✓

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

output.csv

Name homeplot x Find

Ext...	Entry N...	Sequ...	Parent...	Parent...	Update Sequen...	Update Timestamp	Update Reasons	File Attri...
.zip							create	
Name: homeplot-main.zip (Count: 18)								
.zip	30497	16	4460	16	1765809680	2026-04-15 08:51:19	FileCreate	Archive
.zip	30497	16	4460	16	1765809776	2026-04-15 08:51:19	FileCreate SecurityChange	Archive
.zip	30497	16	4460	16	1765809872	2026-04-15 08:51:19		Archive
.zip	30497	16	4460	16	1765809968	2026-04-15 08:51:19		Archive
.zip	30497	16	4460	16	1765810064	2026-04-15 08:51:19		Archive
.zip	30497	16	4460	16	1765810160	2026-04-15 08:51:19		Archive
.zip	30497	16	4460	16	1765810256	2026-04-15 08:51:19		Archive
.zip	30497	16	4460	16	1765810352	2026-04-15 08:51:19		Archive
.zip	30497	16	4460	16	1765810448	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765992848	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765992944	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993040	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993136	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993232	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993328	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993424	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993520	2026-04-15 08:51:19		Archive
.zip	148215	23	4460	16	1765993616	2026-04-15 08:51:19		Archive
Name: homeplot-v1.zip (Count: 9)								
Name: homeplot-v2.zip (Count: 9)								

Format Replace ORF OR LF Comma Tab Font size 12

Extension Contains .zip And Update Reasons Contains create

Total lines 332,257 Visible lines 36 Open files: 1 Search options

The source code from the suspect workstation was archived.
What was the name of the file believed to contain these files? (2 points)

files.zip

Correct! ✓

Based on the question, the file contains these archived files which means the file is a compressed file.

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

output.csv

Enter text to search... Find

Extension	Entry Number	Sequence Number	Parent Entry Number	Parent Sequence Number	Parent Path	Update Sequence Number	Update Timestamp	Update Reasons
zip	=	=	=	=		=	=	
> Name: \$INZU1FU.zip (Count: 14)								
> Name: \$ISGD341.zip (Count: 14)								
> Name: \$IXS1XDO.zip (Count: 14)								
> Name: \$R4SQQU3.zip (Count: 4)								
> Name: \$RAXXS06.zip (Count: 4)								
> Name: \$RGT3I7.zip (Count: 4)								
> Name: \$RNZU1FU.zip (Count: 4)								
> Name: \$RSGD341.zip (Count: 4)								
> Name: \$RXS1XDO.zip (Count: 4)								
> Name: 04-14-2026-13-31-54_files_list.zip (Count: 8)								
> Name: 04-14-2026-14-08-31_files_list.zip (Count: 8)								
> Name: 2026-04-14T141755_Out.zip (Count: 7)								
> Name: ad.zip (Count: 3)								
> Name: AwsEnaNetworkDriver.zip (Count: 1)								
> Name: AWSNVMe.zip (Count: 4)								
> Name: Files.zip (Count: 4)								
> Name: homepilot-main.zip (Count: 32)								
> Name: homepilot-v1.zip (Count: 14)								
> Name: homepilot-v2.zip (Count: 12)								
> Name: kape.zip (Count: 7)								
> Name: NAE-Files.zip (Count: 6)								
> Name: NAE-Files.zip (Count: 5)								
> Name: NAE-Report.zip (Count: 4)								

Extension Contains zip

Total lines 332,257 Visible lines 235 Open files: 1 Search options

searching 'rar', but there is no results. I search '7z', and get 7 items.

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

output.csv

Enter text to search... Find

Extension	Entry Number	Sequence Number	Parent Entry Number	Parent Sequence Number	Parent Path	Update Sequence Number	Update Timestamp	Update Reasons
7z	=	=	=	=		=	=	
> Name: CHROME.7Z (Count: 4)								
> Name: msedge.7z (Count: 5)								
> Name: MSEdge.7z (Count: 8)								
> Name: msedge.hollow.7z (Count: 16)								
> Name: msedge.rebuilt.7z (Count: 4)								
> Name: updater.7z (Count: 12)								
> Name: UPDATER.PACKED.7Z (Count: 8)								

Extension Contains 7z

Total lines 332,257 Visible lines 57 Open files: 1 Search options

Following Q7, when was the archive file created? (1 points)

2026-04-15 10:51:07

Correct! ✓

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

output.csv

Name

Enter text to search... Find

Extension	Entry Number	Sequence Number	Parent Entry Number	Parent Sequence Number	Parent Path	Update Sequence Number	Update Timestamp	Update Reasons
zip	=	=	=	=	=	=	=	=
> Name: \$INZU1FU.zip (Count: 14) > Name: \$ISGD341.zip (Count: 14) > Name: \$IXS1XD0.zip (Count: 14) > Name: \$R4SQQU3.zip (Count: 4) > Name: \$RAXXS06.zip (Count: 4) > Name: \$RGM7317.zip (Count: 4) > Name: \$RNZU1FU.zip (Count: 4) > Name: \$RSGD341.zip (Count: 4) > Name: \$RXS1XD0.zip (Count: 4) > Name: 04-14-2026-13-31-54_files_list.zip (Count: 8) > Name: 04-14-2026-14-08-31_files_list.zip (Count: 8) > Name: 2026-04-14T141755_Out.zip (Count: 7) > Name: ad.zip (Count: 3) > Name: AusEnalNetworkDriver.zip (Count: 1) > Name: AMSNMe.zip (Count: 4) > Name: Files.zip (Count: 4)								
.zip	148282	20	4460	16		1766177240	2026-04-15 10:51:07	FileCreate
.zip	148282	20	4460	16		1766177320	2026-04-15 10:51:07	DataExtend FileCreate
.zip	148282	20	4460	16		1766177400	2026-04-15 10:51:07	DataOverwrite DataExtend FileCreate
.zip	148282	20	4460	16		1766177480	2026-04-15 10:51:07	DataOverwrite DataExtend FileCreate Close
> Name: homepilot-main.zip (Count: 32) > Name: homepilot-v1.zip (Count: 14) > Name: homepilot-v2.zip (Count: 12) > Name: kape.zip (Count: 7)								

☒ Extension
 ☒ Contains
 ☒ zip

C:\Users\BTLOT\Desktop\output.csv

Total lines 332,257 Visible lines 235 Open files: 1 Search options

The user downloaded a browser to cover its tracks. What browser was downloaded? What version? (1 points)

tor, 15.0.9

Correct! ✓

We know the browser downloaded for covering the artifacts, so we can follow the timeline to filter the items after the latest artifacts. A downloaded browser have to be a exe file with 'create' in the Update Reason feild.

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

output.csv

Drag a column header here to group by that column

Enter text to search... Find

Name	Extension	Entry N.	Sequen.	Parent N.	Parent S.	Parent P.	Update Sequ.	Update Timestamp	Update Reasons
\$ISD6MND.exe	.exe	148216	15	41013	10		1765828432	2026-04-15 08:52:33	FileCreate
\$ISD6MND.exe	.exe	148216	15	41013	10		1765828520	2026-04-15 08:52:33	DataExtend FileCreate
\$ISD6MND.exe	.exe	148216	15	41013	10		1765828608	2026-04-15 08:52:33	DataExtend FileCreate Close
\$IQC37PB.exe	.exe	148226	21	41013	10		1765829400	2026-04-15 08:52:41	FileCreate
\$IQC37PB.exe	.exe	148226	21	41013	10		1765829488	2026-04-15 08:52:41	DataExtend FileCreate
\$IQC37PB.exe	.exe	148226	21	41013	10		1765829576	2026-04-15 08:52:41	DataExtend FileCreate Close
tor-browser-windows-x86_64-portable-15.0.9.exe	.exe	148286	24	1904	24		1766188616	2026-04-15 10:52:44	FileCreate
tor-browser-windows-x86_64-portable-15.0.9.exe	.exe	148286	24	1904	24		1766188768	2026-04-15 10:52:44	FileCreate FileDelete Close
firefox.exe	.exe	148315	23	148304	13		1766198528	2026-04-15 10:53:42	FileCreate
firefox.exe	.exe	148315	23	148304	13		1766198616	2026-04-15 10:53:42	DataExtend FileCreate
firefox.exe	.exe	148315	23	148304	13		1766198704	2026-04-15 10:53:42	DataExtend FileCreate BasicInfoChange
firefox.exe	.exe	148315	23	148304	13		1766198792	2026-04-15 10:53:42	DataExtend FileCreate BasicInfoChange Close
nmhproxy.exe	.exe	148332	11	148304	13		1766202280	2026-04-15 10:53:44	FileCreate
nmhproxy.exe	.exe	148332	11	148304	13		1766202368	2026-04-15 10:53:44	DataExtend FileCreate
nmhproxy.exe	.exe	148332	11	148304	13		1766202456	2026-04-15 10:53:44	DataExtend FileCreate BasicInfoChange
nmhproxy.exe	.exe	148332	11	148304	13		1766202544	2026-04-15 10:53:44	DataExtend FileCreate BasicInfoChange Close
plugin-container.exe	.exe	148340	16	148304	13		1766205080	2026-04-15 10:53:47	FileCreate
plugin-container.exe	.exe	148340	16	148304	13		1766205168	2026-04-15 10:53:47	DataExtend FileCreate
plugin-container.exe	.exe	148340	16	148304	13		1766205256	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange
plugin-container.exe	.exe	148340	16	148304	13		1766205344	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange Close
updater.exe	.exe	148348	22	148304	13		1766207352	2026-04-15 10:53:47	FileCreate
updater.exe	.exe	148348	22	148304	13		1766207440	2026-04-15 10:53:47	DataExtend FileCreate
updater.exe	.exe	148348	22	148304	13		1766207528	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange
updater.exe	.exe	148348	22	148304	13		1766207616	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange Close
tor.exe	.exe	148398	15	148397	18		1766218088	2026-04-15 10:53:58	FileCreate
tor.exe	.exe	148398	15	148397	18		1766218176	2026-04-15 10:53:58	DataExtend FileCreate
tor.exe	.exe	148398	15	148397	18		1766218264	2026-04-15 10:53:58	DataExtend FileCreate BasicInfoChange
tor.exe	.exe	148398	15	148397	18		1766218352	2026-04-15 10:53:58	DataExtend FileCreate BasicInfoChange Close
conjure-client.exe	.exe	148400	16	148399	14		1766218616	2026-04-15 10:53:58	FileCreate
conjure-client.exe	.exe	148400	16	148399	14		1766218712	2026-04-15 10:53:59	DataExtend FileCreate

☒ Extension
 ☒ Contains
 ☒ exe
 And
 ☒ Update Reasons
 ☒ Contains
 ☒ create
 And
 ☒ Update Timestamp
 > 2026-04-15 00:00:00

C:\Users\BTLOT\Desktop\output.csv

Total lines 332,257 Visible lines 60 Open files: 1 Search options

Following Q9, when was the browser installer first observed being written to disk, and what entry number corresponds to that file? (2 points)

2026-04-15 10:52:44, 148286

Correct! ✓

Timeline Explorer v3.0.0.1

FileToolsTabsViewHelp

output.csv

Drag a column header here to group by that column

Enter text to search...Find

Name	Extension	Entry No.	Sequence	Parent	Parent S.	Parent	Update Sequence	Update Timestamp	Update Reasons
\$ISD6MMU.exe	.exe	148216	15	41013	10		1765828432	2026-04-15 00:52:33	FileCreate
\$ISD6MMU.exe	.exe	148216	15	41013	10		1765828520	2026-04-15 00:52:33	DataExtend FileCreate
\$ISD6MMU.exe	.exe	148216	15	41013	10		1765828608	2026-04-15 00:52:33	DataExtend FileCreate Close
\$IQC37P8.exe	.exe	148226	21	41013	10		1765829400	2026-04-15 00:52:41	FileCreate
\$IQC37P8.exe	.exe	148226	21	41013	10		1765829488	2026-04-15 00:52:41	DataExtend FileCreate
\$IQC37P8.exe	.exe	148226	21	41013	10		1765829576	2026-04-15 00:52:41	DataExtend FileCreate Close
tor-browser-windows-x86_64-portable-15.0.9.exe	.exe	148286	24	1984	24		1766188616	2026-04-15 10:52:44	FileCreate
tor-browser-windows-x86_64-portable-15.0.9.exe	.exe	148286	24	1984	24		1766188768	2026-04-15 10:52:44	FileCreate FileDelete Close
firefox.exe	.exe	148315	23	148304	13		1766198528	2026-04-15 10:53:42	FileCreate
firefox.exe	.exe	148315	23	148304	13		1766198616	2026-04-15 10:53:42	DataExtend FileCreate
firefox.exe	.exe	148315	23	148304	13		1766198704	2026-04-15 10:53:42	DataExtend FileCreate BasicInfoChange
firefox.exe	.exe	148315	23	148304	13		1766198792	2026-04-15 10:53:42	DataExtend FileCreate BasicInfoChange Close
nhproxy.exe	.exe	148332	11	148304	13		1766202200	2026-04-15 10:53:44	FileCreate
nhproxy.exe	.exe	148332	11	148304	13		1766202368	2026-04-15 10:53:44	DataExtend FileCreate
nhproxy.exe	.exe	148332	11	148304	13		1766202456	2026-04-15 10:53:44	DataExtend FileCreate BasicInfoChange
nhproxy.exe	.exe	148332	11	148304	13		1766202544	2026-04-15 10:53:44	DataExtend FileCreate BasicInfoChange Close
plugin-container.exe	.exe	148340	16	148304	13		1766205080	2026-04-15 10:53:47	FileCreate
plugin-container.exe	.exe	148340	16	148304	13		1766205184	2026-04-15 10:53:47	DataExtend FileCreate
plugin-container.exe	.exe	148340	16	148304	13		1766205288	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange
plugin-container.exe	.exe	148340	16	148304	13		1766205392	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange Close
updater.exe	.exe	148348	22	148304	13		1766207352	2026-04-15 10:53:47	FileCreate
updater.exe	.exe	148348	22	148304	13		1766207488	2026-04-15 10:53:47	DataExtend FileCreate
updater.exe	.exe	148348	22	148304	13		1766207576	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange
updater.exe	.exe	148348	22	148304	13		1766207664	2026-04-15 10:53:47	DataExtend FileCreate BasicInfoChange Close
tor.exe	.exe	148398	15	148397	18		1766218088	2026-04-15 10:53:58	FileCreate
tor.exe	.exe	148398	15	148397	18		1766218168	2026-04-15 10:53:58	DataExtend FileCreate
tor.exe	.exe	148398	15	148397	18		1766218248	2026-04-15 10:53:58	DataExtend FileCreate BasicInfoChange
tor.exe	.exe	148398	15	148397	18		1766218328	2026-04-15 10:53:58	DataExtend FileCreate BasicInfoChange Close
conjure-client.exe	.exe	148400	16	148399	14		1766218616	2026-04-15 10:53:58	FileCreate
conjure-client.exe	.exe	148400	16	148399	14		1766218712	2026-04-15 10:53:59	DataExtend FileCreate

ExtensionContains.exeAndUpdateReasonsContainscreateAndUpdateTimestamp>2026-04-15 00:00:00

Total lines 332/257Visible lines 69Open files: 1Search options